DEBANJAN KAKATI

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EDUCATION

Birla Institute of Technology and Science, Pilani - Goa Campus Aug 2023 - Jul 2027
B.E. Chemical (Hons.) | Minor: Finance

Gurukul Grammar Senior Secondary School 2010 - 2023
Class XII (CBSE): 90% | Class X: 91.6%

SKILLS

- **Languages:** Python (Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, OpenCV), SQL, C++, JavaScript, Java, MATLAB
- **AI & Quant:** LLMs & RAG, Transformers, CNNs, GNNs, LSTM/ARIMA, Explainable AI, AR-GARCH, Monte Carlo, DCF/WACC
- **Tools:** FastAPI, React, LangGraph, Docker, Azure, AWS, **Power BI, Advanced Excel, PostgreSQL, Git, ANSYS, SolidWorks**

RELEVANT COURSEWORK

- Security Analysis & Portfolio Management, Derivatives & Risk Management, Business Analysis & Valuation, Financial Management
- Machine Learning / AI, Deep Learning & Generative AI, Data Analytics, Probability & Stochastic Processes, Plant Design

EXPERIENCE

Digitalization Intern - Innovation & Learning Solutions May 2026 - Dec 2026
Shell India Markets Pvt. Ltd. *Bengaluru, India*

- Architected an end-to-end digital transformation platform for an **8-parallel-reactor** High-Throughput Experimentation (HTE) catalyst testing unit, cutting campaign data assembly from **~10 days to real time**.
- Built automated pipelines ingesting online **Gas Chromatography (GC)** and **SCADA** telemetry, with mass balance closure engines.
- Developed the **DHA Octane Data Merger**, batch-processing analyser reports to automate **RON/MON** and C1–C15 breakdowns.

Business Analytics & Industry 4.0 Intern May 2025 - Jul 2025
PIEDS - BITS Pilani (Entrepreneurship Development Society, Govt. of India) *Remote*

- Evaluated **50+ early-stage startups** on commercial viability, market sizing, unit economics and growth defensibility to guide incubation and **Entrepreneur-in-Residence** decisions.
- Facilitated outreach to **20+ venture capital funds**, securing funding conversations and a network of **50+ mentors**.

Business & Operations Growth Associate Aug 2023 - Jul 2024
INTELLIPAAT *Remote*

- Led development of **3 AI-driven operational analytics systems** (multi-modal decision platform, traffic optimization model, HR automation tool), improving process **efficiency by 30–45%** and cutting manual workload by **50%+**.
- Built end-to-end dataflow pipelines (ingestion → inference → dashboards), **outperforming peer benchmarks by ~35%**.

PROJECTS

Ice Melting Rate Prediction via ML - Patent Filed | Prof. Anirban Roy

- Engineered a patent-pending ML model predicting ice melting dynamics from thermodynamic properties on **237+ experimental data points**, reaching **96% accuracy** (RMSE: 46.8, MAE: 36.4).
- Found a perfect heat-capacity correlation (**r = 1.0**) and benchmarked **12+ regression and deep learning models**, tuning a TensorFlow/Keras network that needed **~40% less training data**.

Systemic Risk in the Cryptocurrency Ecosystem | Prof. Ashwini Kumar

- Built a multi-stage econometric framework combining **AR-GARCH(1,1)**, **Transfer Entropy** spillover networks and leave-one-out systemic indices across **8 digital assets over 10 years**.
- Showed stablecoins USDT (SI 0.1316) and USDC (SI 0.1222) transmit **more systemic contagion than BTC and ETH**, and modelled tail risk with quantile regression (τ = 0.1, 0.5, 0.9).

FreeWind - Spatiotemporal AQI Forecasting | Smart India Hackathon (National Top 1%)

- Designed an air-quality forecasting engine fusing **Graph Neural Networks** and **bidirectional LSTMs** across **300,000+ multi-city sensor records**, achieving **32% lower MAE** than traditional baselines.
- Engineered real-time pollution source attribution fusing satellite imagery, traffic heatmaps and meteorological telemetry.

Paints Industry Equity Valuation & Financial Modelling | Prof. Mohit Kumar

- Built **3-statement models** for Asian Paints, Berger Paints and Kansai Nerolac, projecting 5-year FCFF with terminal value.
- Estimated Cost of Equity at **13.21%** (CAPM), **WACC of 11.8%–12.4%**, ±100 bps sensitivity and EV/EBITDA comps (**35.44x**).

POSITION OF RESPONSIBILITY

SAE Formula Bharat - Chassis Development Engineer

- Engineered **7+ optimized CAD/FEA designs** (ANSYS, SolidWorks), contributing to a **Top 25 national placement among 80+ teams**.
- Accelerated vehicle completion by **2+ weeks through early FEA validation** while leading chassis integration.

PATENT & CERTIFICATIONS

PATENT: Ice Melting Rate Prediction Model Filed: Jan 27, 2025 | Issued: Nov 5, 2025 | App No. IDF/H/115107 | India
CERTIFICATION: Advanced Data Science & Artificial Intelligence | Intellipaat **2023**
CERTIFICATION: Advanced Artificial Intelligence & Robotics | myEquation - ROBO AI **2024**

ACHIEVEMENTS

- **Indian Space Academy** - Remote Sensing & GIS (Winter Training): certified in geospatial analysis and satellite data interpretation.